

Creative Innovation Call 2026

Projectplan & Begroting | Project Plan & Budget

Health-Data-Design-Playground (HDDP):

Innovating Data-Driven Everyday Health Interventions

Summary (194words / 200 max.)

“What if health data didn’t just live in spreadsheets, but came to life through interactive objects and immersive spaces around us?” The **Health-Data-Design-Playground (HDDP)** turns this vision into reality by empowering designers to transform raw sensor data into engaging, data-driven everyday health interventions through hands-on experimentation with health sensors and personal data. While sensing and health-monitoring technologies now provide increasingly accurate insights, many current health interventions fail to achieve long-term impact due to poor integration into daily life and a lack of compelling interaction and experiential design. HDDP bridges the gap between design intent and technical implementation. By offering accessible sensor toolkits, shared datasets, and design-oriented software resources, the platform lowers the entry barrier to working with complex biosensors and data pipelines, enabling designers to move beyond speculative concepts toward rapid, real-world prototyping, supporting the creation of feasible, experiential, and human-centered health solutions. As a shared educational and innovation infrastructure within HvA, HDDP supports interdisciplinary education and collaboration across design, health, and technology programs. In the long term, HDDP aims to foster meaningful, engaging, and sustainable health interventions that encourage personal reflection, support behavior change, and strengthen participatory healthcare alongside traditional clinical care.

1. Relevance (783words / 800 max.)

1.1 Background

With the rise of participatory healthcare [1], individuals are increasingly expected to take an active role in managing their own health [2]. Establishing healthy habits, such as regular physical activity and sufficient sleep, is now vital for long-term well-being. With the maturation of sensing technologies and health-monitoring devices, personal health data is increasingly accessible and serves as the foundation for data-driven interventions to enhance self-management and promote healthful behaviors [3-5]. However, while these interventions provide accurate data and personalized health insights, their impact is often limited by their form-giving and their lack of integration into daily life [6-7]. Addressing these constraints requires more than just technical advancement; it calls for design practitioners to actively engage in crafting new forms of interventions that are compelling, motivating, and capable of sustaining long-term use [8,9].

1.2 Current challenges

Hands-on experimentation with sensing techniques and real-user data is a critical prerequisite for creating meaningful, data-driven health interventions [10]. However, design professionals, especially those with non-technical backgrounds, often face significant barriers when integrating sensing technologies, electronic hardware, and data-processing algorithms into their designs [11]. The technical complexity requires specialized expertise and significant time investment, which frequently hinders early-stage design exploration and confines designers to conceptual or speculative work. As a result, a critical gap emerges between design intent and practical implementation, limiting designers’ ability to assess technical feasibility, improve interaction quality, implement, and evaluate their impact in a real-world setting.

Centre of Expertise Creative Innovation

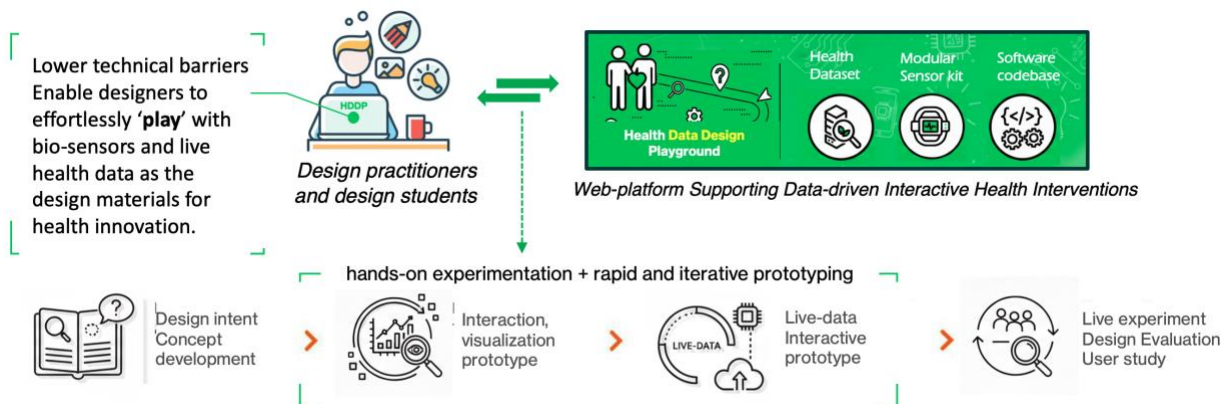


Fig 1. Health-Data-Design-Playground (HDDP) aims to support more design practitioners in healthcare innovations

1.3 Project idea and Significance

Through this project, we aim to develop a design platform that enables design practitioners and design students to fast prototype and iterate data-driven interactive health interventions through hands-on experience with health sensors and data. The proposed **Health-Data-Design-Playground (HDDP)** will lower the technical barriers to biosensors and health data processing, enabling designers to more effectively shape innovative, effective, and engaging health interventions. This shift empowers designers to contribute more to the transition from treatment-centered healthcare models toward participatory healthcare.

1.4 Intended impact

Firstly, HDDP will support multiple educational programs at the HvA. Taking our current collaboration with FDMCI's educational programs as an example, HDDP could support:

- Minor Information Design CDM, where students require accessible datasets to explore data visualization and data-driven storytelling
- Minor Immersive Environments, where students frequently work with sensors to create interactive and embodied experiences.
- Master Digital Design, where students engage with data-driven systems and interactive prototyping.
- Master Applied AI, where students leverage AI and data to develop smart health systems.

Beyond FDMCI, HDDP could also support graduation students' projects in the broader context of the HvA, for example, in the Faculty of Health, Sports, and Exercise, and the Faculty of Technology. By providing shared datasets, sensor toolkits, and design-oriented software resources, the HDDP will allow students to move more quickly from concept to functional prototype, while developing a deeper understanding of health data, interaction design, and experiential systems.

Secondly, HDDP will support design practitioners and SMEs in engaging with health innovation by lowering technical barriers and providing practical entry points for designing interactive, data-driven health interventions. In our prior collaborations, we partnered with design studios on projects such as HeartBloom [12], a metaphorical stress visualization, and BioMirror [13], a tangible biofeedback interface. By providing technical support for sensing technologies and data processing, these projects enabled those designers to focus on interface design, aesthetics, and user experience. These outcomes demonstrate that access to sensor and data resources is critical for translating design intent into meaningful health interventions. The proposed HDDP, offering accessible sensor toolkits and datasets, will empower more design practitioners to create innovative health-promotion solutions.

Third, in the long term, HDDP aims to cultivate the development of innovative data-driven health interventions that support personal reflection, behavior change, and more informed dialogue between individuals and healthcare professionals, thereby strengthening individuals' active role in managing their own health. Collectively, these effective everyday health interventions can accelerate the shift toward participatory healthcare alongside clinic-centered care and help reduce the burden on the healthcare system in the Netherlands.

1.5 Alignment with the CoE CI programme lines

This project aligns with the thematic line **Media, Technology & Society** by building a bridge that enables designers to effectively leverage data and sensing technologies and to confidently enter the health design field, actively contributing to participatory healthcare innovation. The project focuses on developing a **designer-facing Health-Data-Design-Playground** that provides datasets, toolkits, and design resources to support designers with diverse backgrounds and levels of technical expertise in exploring new forms of human–data interaction in everyday health contexts while addressing key challenges such as meaningful data interpretation, user motivation, and long-term user experience and engagement in participatory healthcare.

2. Plan of Approach (785words / 800 max.)

2.1 Project focus (What we will do)

Through this project, we aim to develop **HvA Health-Design-Data-Playground**, where design practitioners and students, the next generation of designers, can effortlessly 'play' with biosensors (heart rate, breathing, muscle activity) and health data (fitness, sleep, chronic stress) as creative materials for rapid prototyping and hands-on exploration. This will empower them to create more effective, engaging health interventions integrated into everyday life. HvA-HDDP will be built around three core pillars:

1. **Curated health datasets** – Includes sleep, activity, Heart Rate Variability (HRV)/stress, and fitness data sourced from online databases or newly collected via sensors. Data will be cleaned, pre-processed, structured in design-friendly formats (e.g., JSON), and accompanied by clear usage instructions.
2. **Modular sensor kits** – Focused on stress, breathing, and physical activity/posture. Existing devices (e.g., pulse sensors) will be integrated with open-source hardware (Arduino, Raspberry Pi) and outputs like LEDs or actuators, enabling designers to create interactive experiences without advanced technical skills.
3. **Reusable software resources** – Provides kits and code examples in Processing, TouchDesigner, and web-based platforms. Supports commercial sensor APIs (Fitbit, Oura, Polar) and real-time open-source hardware data, allowing designers to quickly build interactive health applications and visualizations.

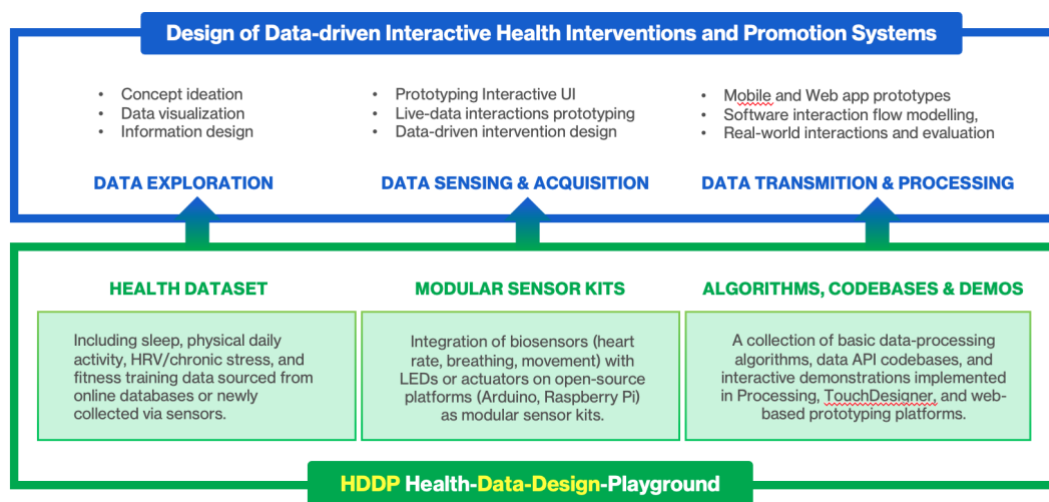


Fig 2. Three Pillars of HDDP: supporting hands-on design of data-driven health interventions

2.2 Planned Activities (How we will do)

Phase 1: Planning & Requirements (Mar-Apr 2026). In Phase 1, we will co-design the HDDP with design practitioners, design lecturers in FDMCI, and master's students from the Interaction Design Track, to identify key technical challenges, resource needs, and preferred tools and platforms for design and prototyping. In parallel, we will review and select commonly available health sensors (e.g., heart rate, breathing) and identify relevant commercial APIs for integration (e.g., Fitbit, Oura Ring, Polar LOOP). Based on this analysis, we will define supported data types, data formats, and data-cleaning standards for curated datasets. **Deliverables:** 1) User requirements and platform specification document. 2) List of selected sensors and APIs. 3) Data management and processing plan.

Phase 2: Dataset Collection & Preparation (May-Jun 2026). Phase 2 focuses on curating and preparing health-related existing datasets from publicly available sources such as PhysioNet, Kaggle, and Open Humans. All datasets will be cleaned, preprocessed, and structured in design-friendly formats (e.g., JSON). The phase will conclude with the development of comprehensive documentation to support designers in understanding, interpreting, and applying the datasets in design projects. **Deliverables:** 1) Curated, cleaned, and pre-processed datasets on sleep, fitness, HRV/stress. 2) Data documentation and user guide.

Phase 3: Modular Data Sensing Kit Development (Jul-Nov 2026). Phase 3 focuses on developing data-sensing kits consisting of integrated hardware and software modules that enable data transmission from sensors to design software or tools. We will integrate three types of sensors (e.g., heart rate, breathing, motion, activity, and sleep sensors) into hardware modules using Arduino and Raspberry Pi. In parallel, we will develop corresponding software modules in *Processing*, *TouchDesigner*, and web-based environments to support real-time sensor data acquisition, visualization, and interactive experiences. **Deliverables:** 1) Modular data-sensing kits (hardware + software). 2) User guides and setup instructions for students and designers

Phase 4: Dissemination & Integration (Dec 26 - Feb 27). In the final phase, we will integrate the curated datasets, sensing kits, and web-based software into a unified online platform. We will conduct pilot studies with design practitioners, design educators, and design students to evaluate usability, functionality, and workflow integration. We will also seek opportunities for integration of the HDDP into educational programmes as mentioned earlier. We are already discussing potential ideas with CMD and the Master Digital Design. **Deliverables:** 1) Fully integrated online platform. 2) Workshop and webinar reports. 3) Public demo of the platform and final user feedback.

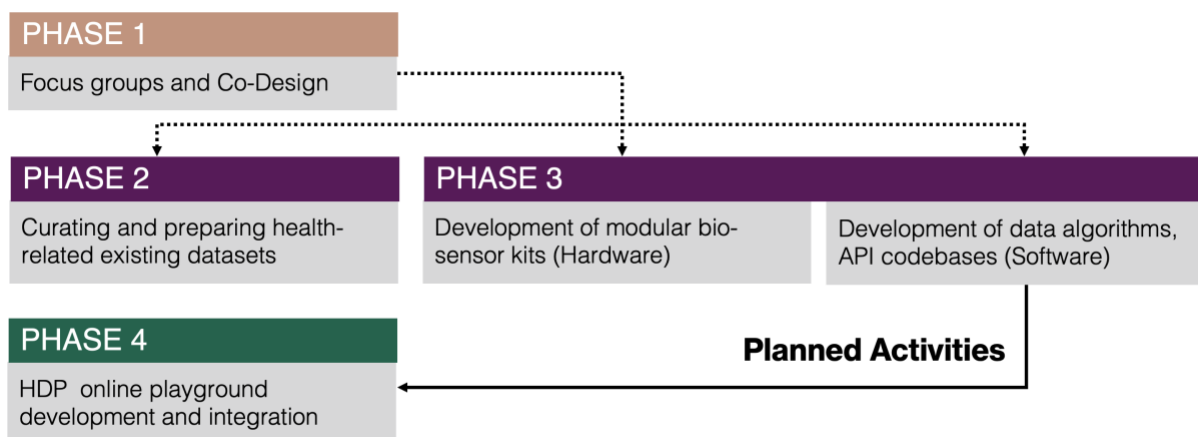


Fig 3. Work plan and activities for co-design and building HDDP

2.3 Intended Outcomes

Based on the deliverables outlined above, this project is expected to achieve **three main outcomes**:

1. **Health-Design-Data-Playground Website:** We will develop a HvA-hosted platform providing ready-to-use datasets, modular sensor kits, and an open-source web codebase, enabling designers and students to rapidly prototype with data-driven health design.
2. **New NWO RAAK MKB Research Proposal:** We will develop a new proposal based on this project, with an extended collaboration with six design SMEs and studios to develop case studies using HDDP, collecting feedback to further refine and enrich the playground.

2.4 Collaboration and Partnerships

This project will foster and strengthen collaboration between the Digital Life Research Group, various educational programs, design studios, and external researchers. The HDDP will be co-designed with these collaborative partners to ensure it is useful for design practitioners, educators, and students. By providing sensor toolkits, sample datasets, and software modules, HDDP will enable partners to co-create and prototype data-driven interactive health interventions.

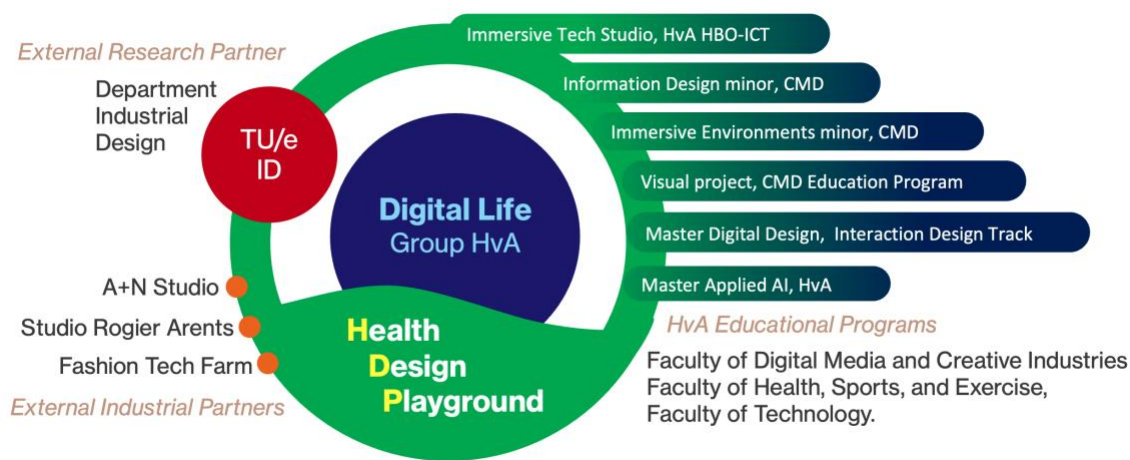


Fig 4. Co-Design the Health-Data-Design-Playground with our collaboration with design researchers (TU/e), educators and students (HvA, CMD, HBO-ICT), design professionals (including A+N Studio [14], Studio Roger Arents [15], Fashion Tech Farm [16]).

3 Begroting | Budget

	Inzet (aantal uren)	Totaal bedrag	Eigen bijdrage	Aangevraagd bedrag	Kosten- plaats nummer
Personeelskosten					
Junior researcher (Design Technologist)	250	14919		14919	103804
Bin Yu	60	7140	7140	0	103804
Somaya Ben Allouch	10	1510	1510	0	103804
Materiële kosten					
Sensors and devices cost		5000		5000	103804
Vergoeding externe partner (incl BTW)					
Totaal		28569	8650	19919	103804
Aangevraagd bedrag					

4. References

1. Donetto, S., Pierri, P., Tsianakas, V., & Robert, G. (2015). Experience-based co-design and healthcare improvement: realizing participatory design in the public sector. *The Design Journal*, 18(2), 227-248.
2. Lu, T., Lin, Q., Yu, B., & Hu, J. (2025). A systematic review of strategies in digital technologies for motivating adherence to chronic illness self-care. *npj Health Systems*, 2(1), 13.
3. Marsch, L. A. (2021). Digital health data-driven approaches to understand human behavior. *Neuropsychopharmacology*, 46(1), 191-196.
4. Burgermaster, M., Desai, P. M., Heitkemper, E. M., Juul, F., Mitchell, E. G., Turchioe, M., ... & Mamykina, L. (2023). Who needs what (features) when? Personalizing engagement with data-driven self-management to improve health equity. *Journal of Biomedical Informatics*, 144, 104419.
5. Nakamura, Y. (2021, September). lot nudge: lot data-driven nudging for health behavior change. In *Adjunct Proceedings of the 2021 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2021 ACM International Symposium on Wearable Computers* (pp. 51-53).
6. Thomas Craig, K. J., Morgan, L. C., Chen, C. H., Michie, S., Fusco, N., Snowdon, J. L., ... & Sill, S. (2021). Systematic review of context-aware digital behavior change interventions to improve health. *Translational behavioral medicine*, 11(5), 1037-1048.
7. Campbell, M., Fitzpatrick, R., Haines, A., Kinmonth, A. L., Sandercock, P., Spiegelhalter, D., & Tyrer, P. (2000). Framework for design and evaluation of complex interventions to improve health. *Bmj*, 321(7262), 694-696.
8. Scholten, H., & Granic, I. (2019). Use of the principles of design thinking to address limitations of digital mental health interventions for youth. *Journal of Medical Internet Research*, 21(1), e11528.
9. Wray, T. B., Kahler, C. W., Simpanen, E. M., & Operario, D. (2019). User-centered, interaction design research approaches to inform the development of health risk behavior intervention technologies. *Internet interventions*, 15, 1-9.
10. Xu, J., Chen, P. Y., Uglow, S., Scott, A., & Montague, E. (2012). Design challenges and guidelines for persuasive technologies that facilitate healthy lifestyles. *Computer technology and application*, 3(2), 140.

Centre of Expertise Creative Innovation

11. Marcu, G., Ondersma, S. J., Spiller, A. N., Broderick, B. M., Kadri, R., & Buis, L. R. (2022). Barriers and considerations in the design and implementation of digital behavioral interventions: qualitative analysis. *Journal of medical Internet research*, 24(3), e34301.
12. Yu, B., & Arents, R. (2020). Biofeedback painting: Let the heart lead the brush. *Leonardo*, 53(5), 504-509.
13. Yu, B., Bongers, N., Van Asseldonk, A., Hu, J., Funk, M., & Feijs, L. (2016, February). LivingSurface: biofeedback through shape-changing display. In *Proceedings of the TEI'16: Tenth International Conference on Tangible, Embedded, and Embodied Interaction* (pp. 168-175).
14. A+N Studio (2026). Retrieved from: <https://www.a-plus-n.com/>
15. Studio Rogier Arents (2026). Retrieved from: <https://www.rogierarents.nl/>
16. Fashion Tech Farm (2026). Retrieved from: <https://fashiontechfarm.com/>